Parsing Documents at Scale

System design diagrams

Discussion of November 12, 2026

1. End-to-end architecture
Source SaaS to cited answers: ingestion, parsing, chunking, indexing, retrieval, evaluation.

2. Ingestion and change detection
Fetcher loop with token bucket, cursor table, S3 versioning, and SQS events.

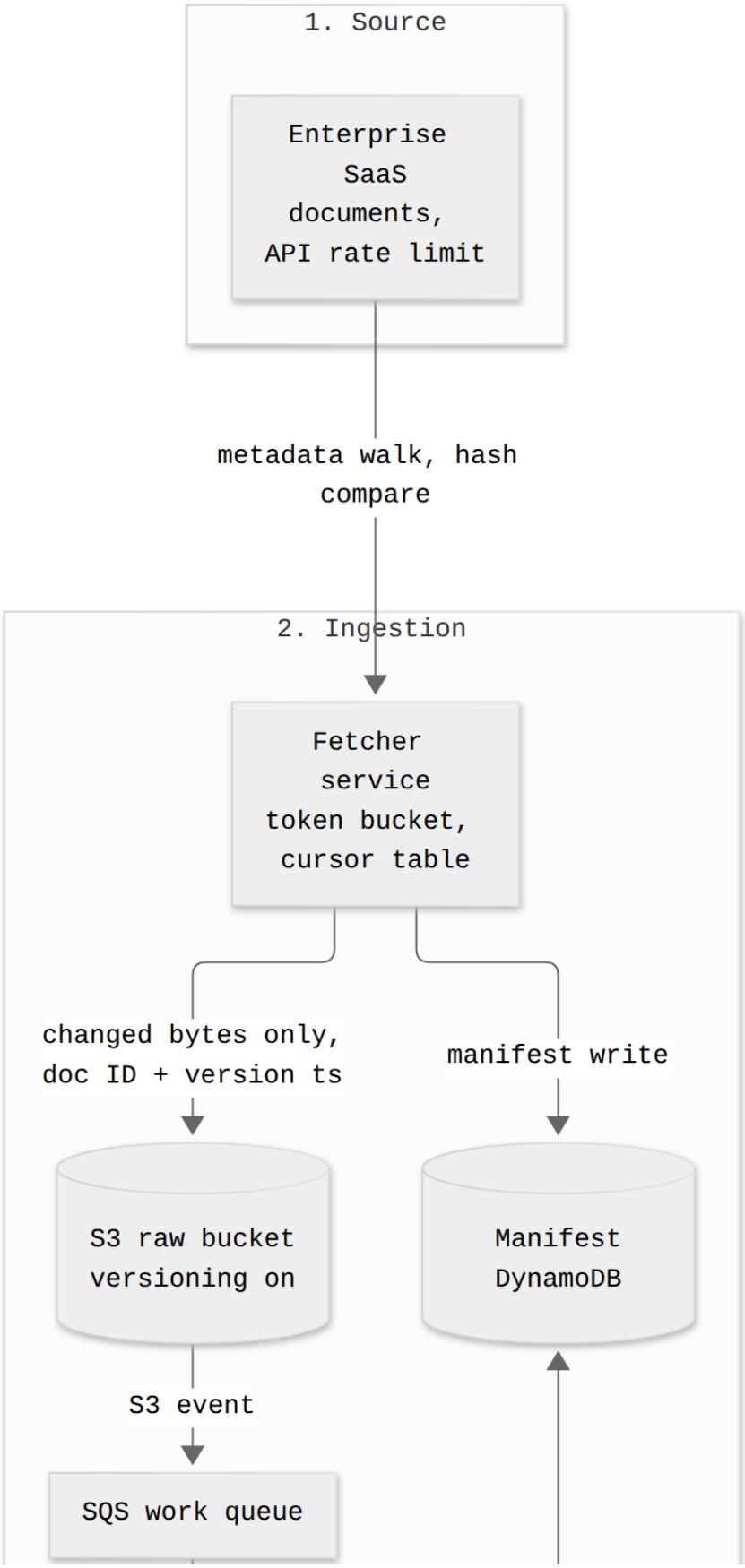
3. Query flow (hybrid retrieval)
Query embedding, vector search and keyword search in parallel, score blend, cited answer.

4. Evaluation loop per stage
Metrics at each stage, scheduled eval runs, CloudWatch dashboard, feedback into the pipeline.

Full write-up: system-design-doc-parsing.md

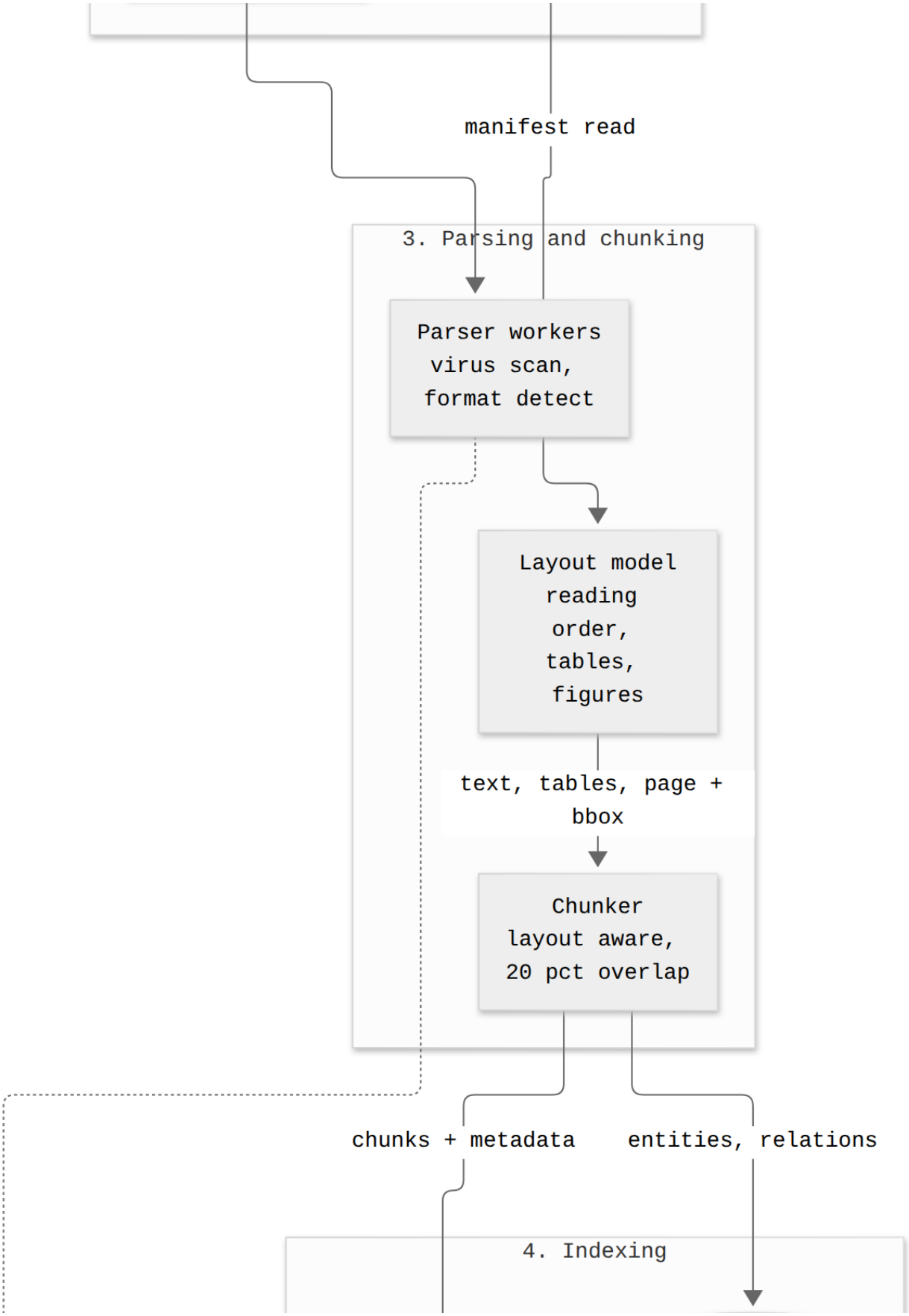
1. End-to-end architecture (page 1 of 5)

Source SaaS to cited answers: ingestion, parsing, chunking, indexing, retrieval, evaluation.



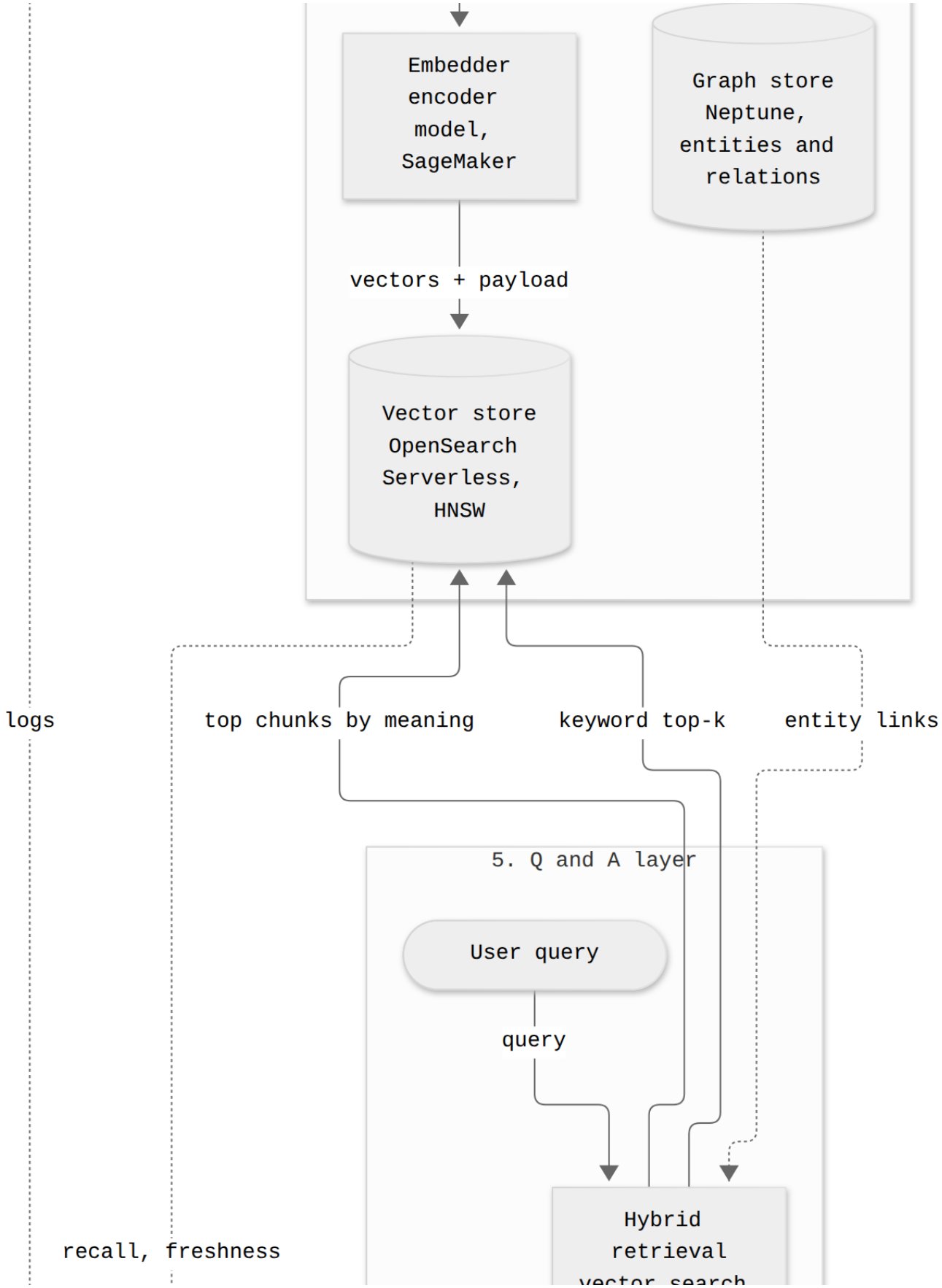
1. End-to-end architecture (page 2 of 5)

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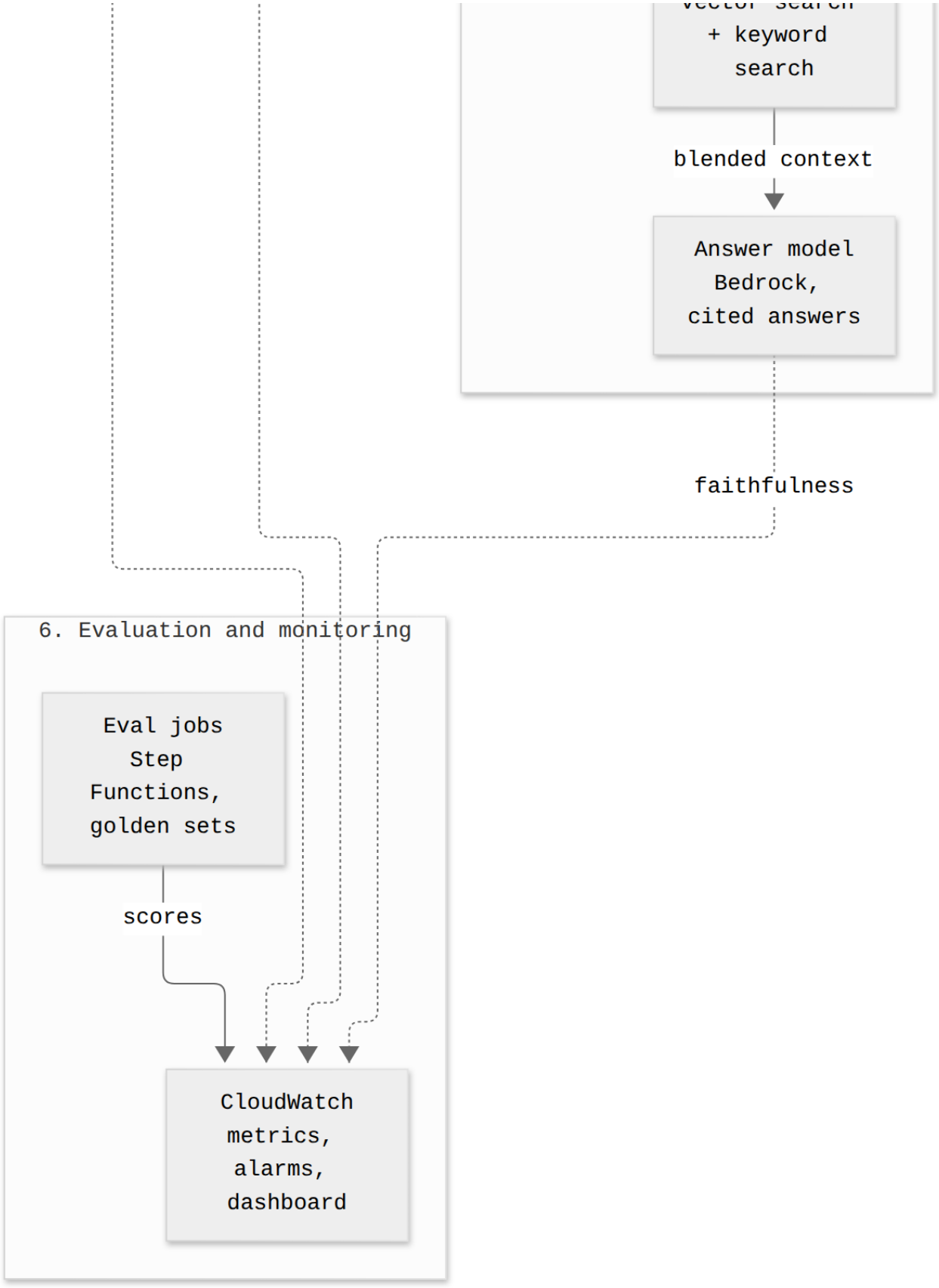
1. End-to-end architecture (page 3 of 5)

Source SaaS to cited answers: ingestion, parsing, chunking, indexing, retrieval, evaluation.



1. End-to-end architecture (page 4 of 5)

Source SaaS to cited answers: ingestion, parsing, chunking, indexing, retrieval, evaluation.

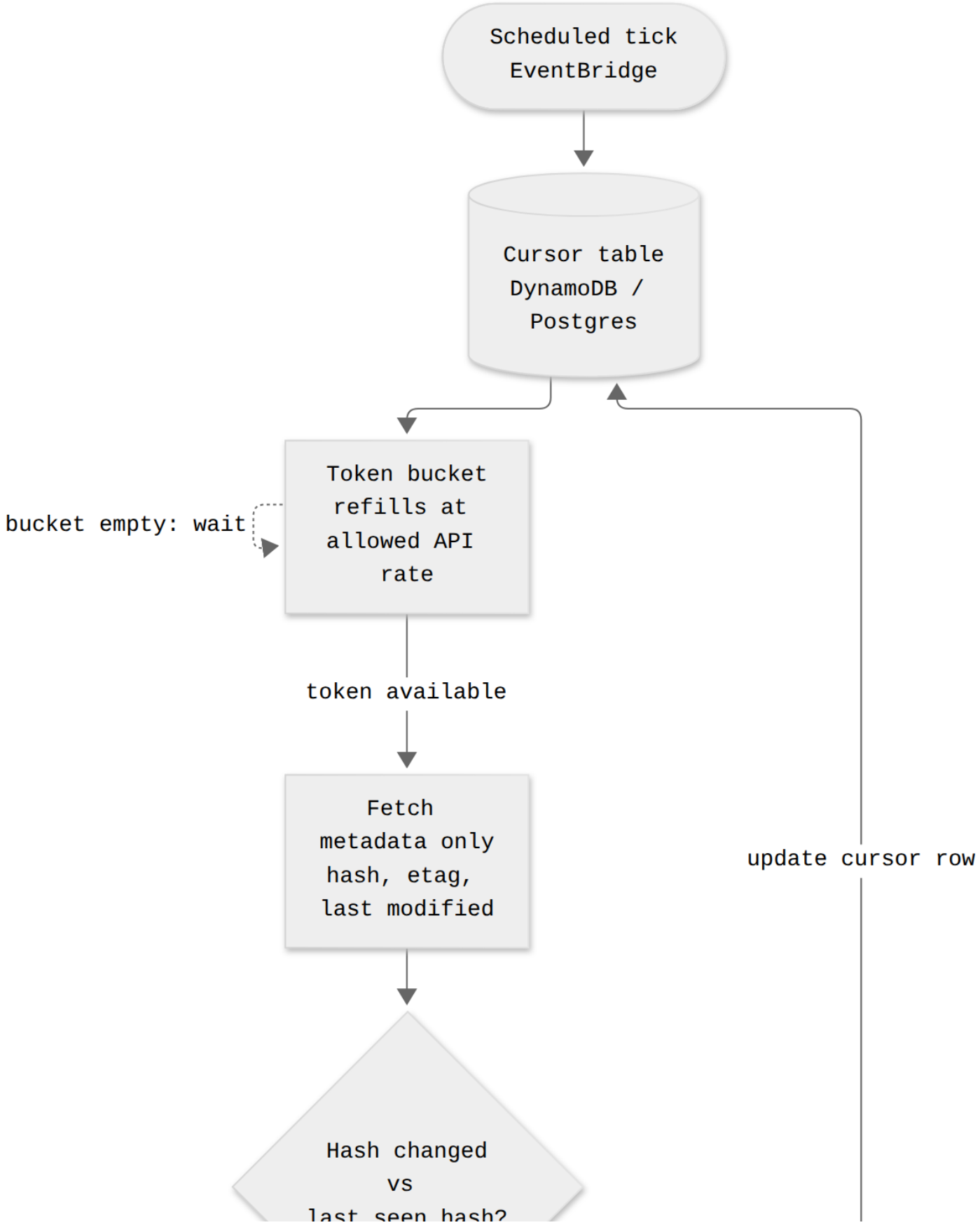


1. End-to-end architecture (page 5 of 5)

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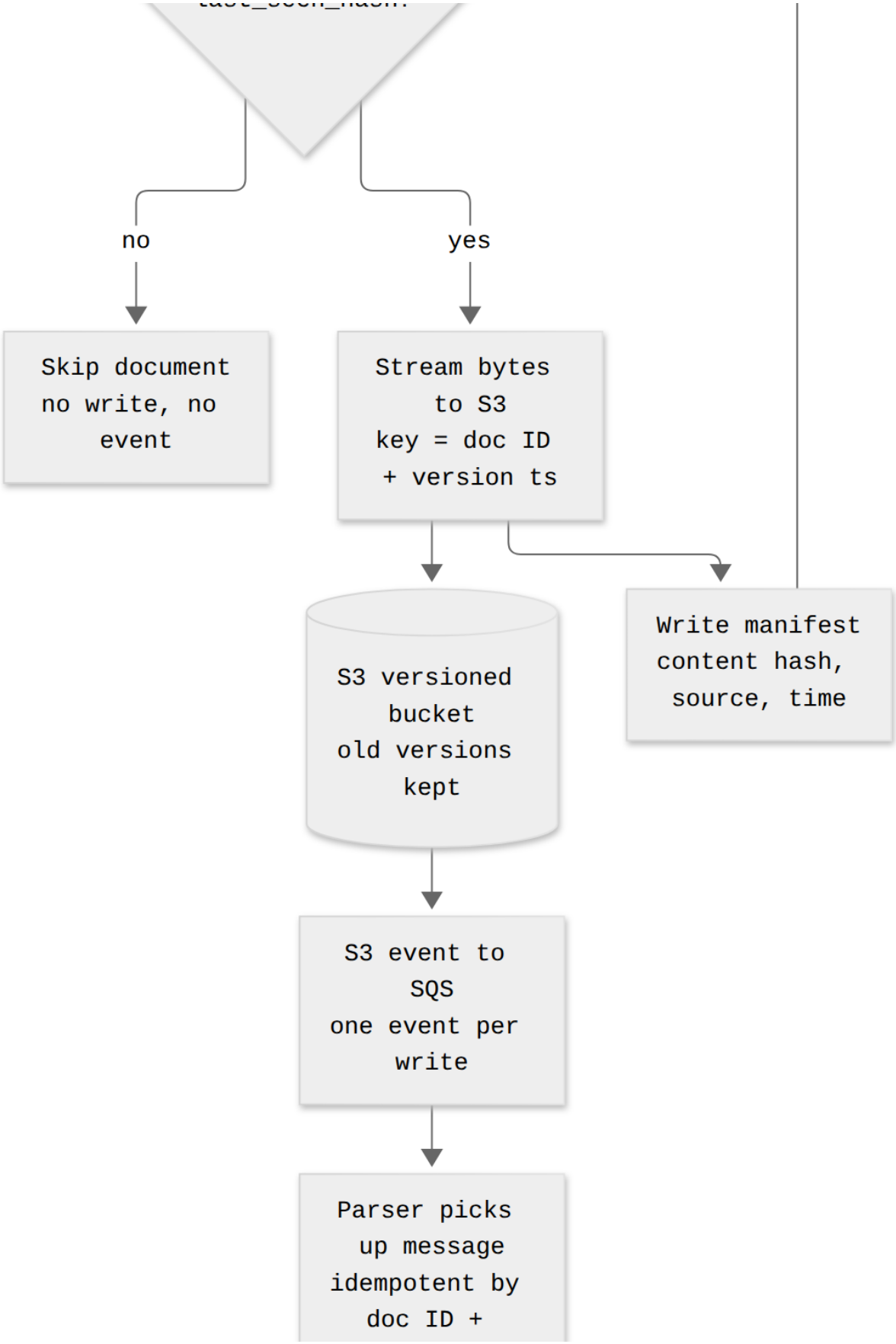
2. Ingestion and change detection (page 1 of 3)

Fetcher loop with token bucket, cursor table, S3 versioning, and SQS events.



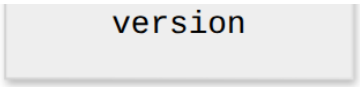
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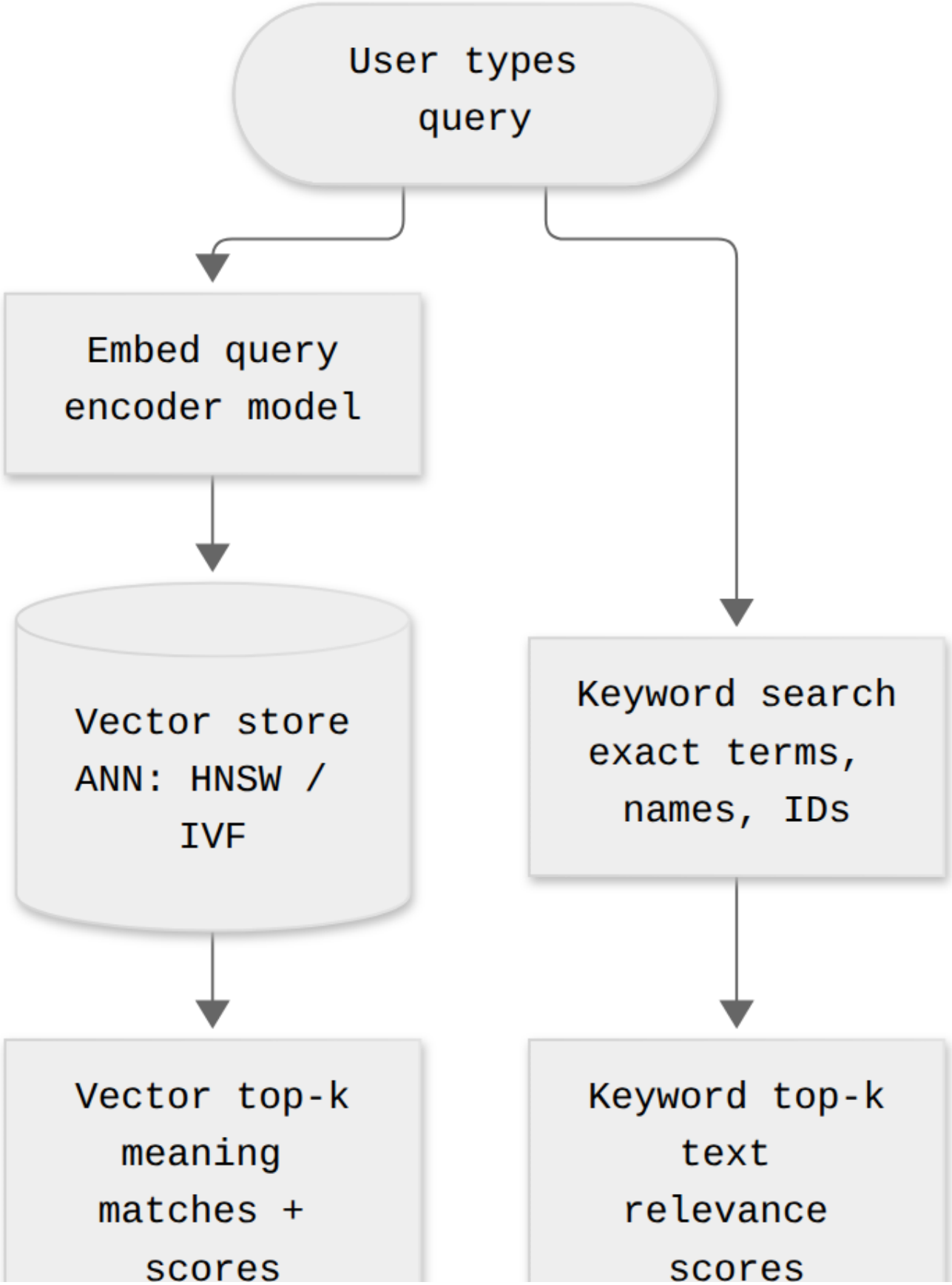
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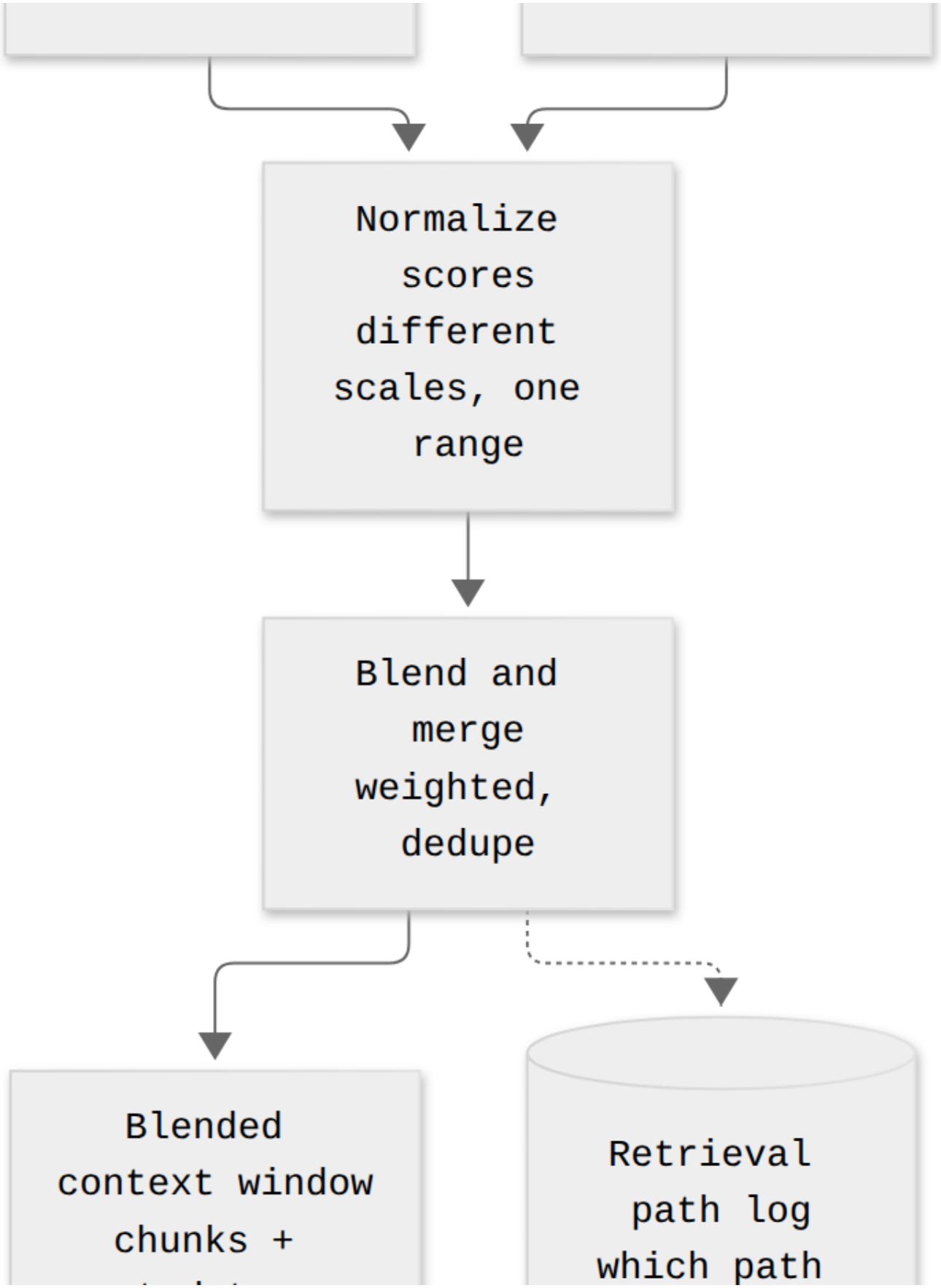
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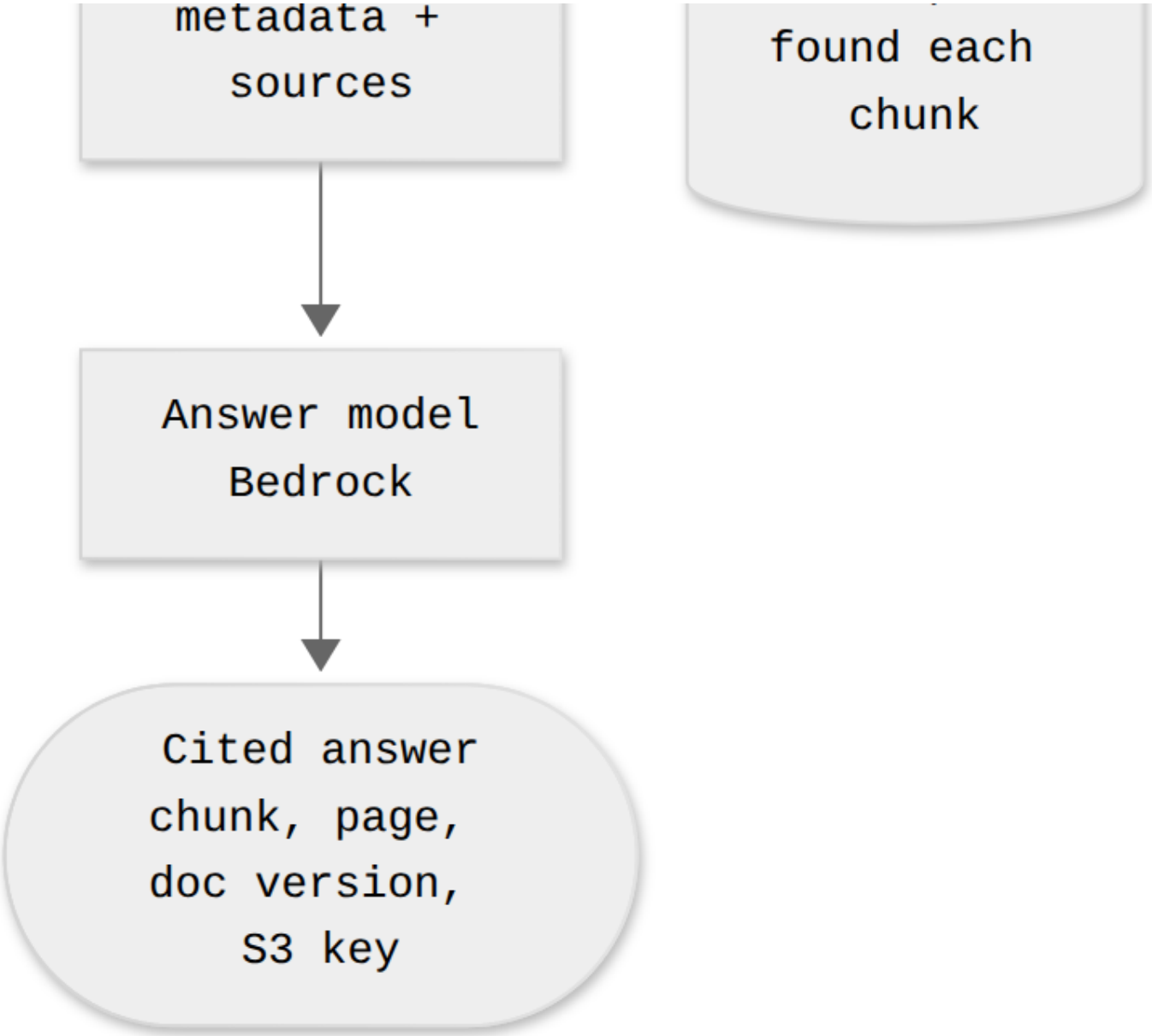
3. Query flow (hybrid retrieval) (page 2 of 3)

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